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US Citizen

Michael Herman

AI/ML & CV Researcher

Google Scholar · ORCID · GitHub · Hugging Face · Zenodo · Tableau Public · LinkedIn · Portfolio

AI and computer vision researcher with 5+ years of research experience developing ML models for satellite-based sensor systems. Specialized in deep learning and state estimation algorithms. Demonstrated success publishing novel space sensor DL models, large-scale image datasets, and a comprehensive sensor algorithm survey. Passionate about advancing AI for defense applications.

Education

- **Georgia Institute of Technology** — Atlanta, GA
  Ph.D. in Aerospace Engineering; GPA: 3.58 — *Jan 2018 - Dec 2024*
- **Georgia Institute of Technology** — Atlanta, GA
  M.S. in Aerospace Engineering; GPA: 3.58 — *Jan 2015 - Dec 2017*
- **Georgia Institute of Technology** — Atlanta, GA
  B.S. in Aerospace Engineering; GPA: 3.61 — *Aug 2012 - Dec 2014*
  Honors: *Highest Honors*

Skills Summary

- **Languages** Python, MATLAB, R, JAVA, C, FORTRAN, PowerShell
- **Frameworks** PyTorch, TensorFlow
- **Libraries** iNNvestigate, Matplotlib, NumPy, OpenCV, Pandas, Scikit-learn, SciPy, Seaborn, SpConv, Tensorboard, Torchvision
- **Tools** Anaconda, Docker, GIT, Jupyter, LaTeX
- **Software** NI LabVIEW, Ansys Zemax OpticStudio, MATLAB, Simulink, STK, Tableau, Unreal Engine 5

Professional Experience

- **Georgia Institute of Technology** — Atlanta, GA
  IT Support Engineer Senior - College of Engineering — *Feb 2025 - Current*
  - Worked with the Georgia Tech AI Makerspace to assist users in developing Artificial Intelligence (AI) tools and managing High-Performance Computing (HPC) GPU clusters for the Partnership for an Advanced Computing Environment (PACE).
  - Developed custom scripts to support a modern SCCM-based Windows imaging workflow, replacing the legacy MDT workflow across campus.
  - Collaborated with IT staff and researchers across the College of Engineering units to provide hardware and software technical support.
- **Georgia Institute of Technology** — Atlanta, GA
  Graduate Teaching Assistant - Aerospace Engineering IT Support — *Aug 2019 - Dec 2024*
  - Provided on-site support for students, faculty and staff in the Aerospace Engineering department to troubleshoot software and hardware.
- **Georgia Institute of Technology** — Atlanta, GA
  Graduate Teaching Assistant - Space Systems Design Laboratory (SSDL) — *Jan 2015 - Dec 2015*
  - Taught, graded and made assignments for the senior design class of the undergraduate space systems track.

Research Experience

- **Georgia Institute of Technology** — Atlanta, GA
  Graduate Research Assistant - Aerospace Systems Design Laboratory (ASDL) — *Aug 2019 - Dec 2024*
  - Prototyped and published sparse submanifold convolutional neural network (SSCNN) for sun sensor attitude determination to HuggingFace.
  - Applied data fusion to a Multiple-Input Multiple-Output (MIMO) deep neural network (DNN) for multi-sensor input and two-axis state output.
  - Developed a physics-based diffraction simulation for model training and testing using OpticStudio and MATLAB via ZOS-API to Github.
  - Created and publicly released a large-scale labeled image dataset (2M+ images) for digital sun sensor model training, hosted on Zenodo/HF.
  - Quantified and analyzed the performance and robustness of the algorithm via error histograms and heatmaps using the Seaborn library.
  - Estimated uncertainty of the deep learning model with Bayesian inference using Monte Carlo Dropout (MCDO).
  - Applied saliency mapping to assess AI explainability of the output image space using the iNNvestigate library.
- **Georgia Institute of Technology** — Atlanta, GA
  Graduate Research Assistant - Space Systems Design Laboratory (SSDL) — *Jan 2016 - Dec 2017*
  - Developed and documented algorithms for an analog sun sensor in the Ranging And Nanosatellite Guidance Experiment (RANGE) mission.
  - Designed an ADCS CubeSat test platform with spherical air bearing test-bed incorporated within a Helmholtz cage for in-house testing.
  - Implemented an Extended Kalman Filter (EKF) within the CubeSat flight control system to estimate and correct analog sun sensor biases.

Projects

- **Dissertation:** Herman, M. (2025). Predictive calibration for digital sun sensors using sparse submanifold convolutional neural networks.
- **Masters Project:** Herman, M. (2016). Design and Application of a Circular Aperture Sun Sensor.

Selected Publications

- Herman, M., Pinon Fischer, O., & Mavris, D. N. (2025). Sun sensor calibration algorithms: A systematic mapping and survey [under-review].
- Herman, M., Pinon Fischer, O., & Mavris, D. N. (2025). Sun sensor calibration algorithms: Survey dataset (1.0) [Data set]. Zenodo.
- Herman, M., Pinon Fischer, O., & Mavris, D. N. (2025). Predictive calibration for digital sun sensors using sparse submanifold convolutional neural networks [under-review].
- Herman, M., Pinon Fischer, O., & Mavris, D. N. (2025). DSS-CNN: calibration image dataset (1.0) [Data set]. Zenodo.
- Herman, M., Pinon Fischer, O., & Mavris, D. N. (2025). DSS-CNN: calibration models (1.0) [Model]. Zenodo.